KAIZEN MATHS

Lesson starter

Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule

Euler Method

Name

S1

Use one Euler step with $y_{n+1} = y_n + hf(x_n,y_n)$.

Use one Euler step with $\frac{dy}{dx} = 1x + y$, $x_0 = 0$, $y_0 = 1$, and $h = 0.5$.

S2

Complete the Euler approximations step by step.

Use Euler's method with $\frac{dy}{dx} = 2x + y$, $y(0) = 1$, and $h = 0.2$ to estimate $y(0.6)$.

S3

Use the Euler update to find the missing step size.

An Euler step gives $y_1 = 1.5$ from $y_0 = 1$ when $f(x_0, y_0) = 2$. Find h.

S4

Use one Euler step with $y_{n+1} = y_n + hf(x_n,y_n)$.

Use one Euler step with $\frac{dy}{dx} = 3x + y$, $x_0 = 0$, $y_0 = 2$, and $h = 0.5$.

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Use one Euler step with $y_{n+1} = y_n + hf(x_n,y_n)$.

Use one Euler step with $\frac{dy}{dx} = 3x + y$, $x_0 = 0$, $y_0 = 2$, and $h = 0.5$.

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KAIZEN MATHS

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Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule

Euler Method

Name _____

EA1

Complete the Euler approximations step by step.

Use Euler's method with $\frac{dy}{dx}$ = 1x + y, y(0) = 1, and h = 0.25 to estimate y(0.75).

EA2

Use the Euler update to find the missing step size.

An Euler step gives y_1 = 4 from y_0 = 2 when f(x_0, y_0) = 8. Find h.

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Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule

Euler Method

Name _____

EB1

Use one Euler step with y_{n+1} = y_n + hf(x_n,y_n).

Use one Euler step with $\frac{dy}{dx}$ = 2x + y, x_0 = 0, y_0 = 1, and h = 0.25.

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Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule

Euler Method

Name _____

EA1

Complete the Euler approximations step by step.

Use Euler's method with $\frac{dy}{dx}$ = 1x + y, y(0) = 1, and h = 0.25 to estimate y(0.75).

EA2

Use the Euler update to find the missing step size.

An Euler step gives y_1 = 4 from y_0 = 2 when f(x_0, y_0) = 8. Find h.

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Numerical Methods: Iteration, Newton-Raphson and Trapezium Rule

Euler Method

Name _____

EB1

Use one Euler step with y_{n+1} = y_n + hf(x_n,y_n).

Use one Euler step with $\frac{dy}{dx}$ = 2x + y, x_0 = 0, y_0 = 1, and h = 0.25.

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Teacher answers & working

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S1

Use one Euler step with $y_{n+1} = y_n + hf(x_n,y_n)$.

Use one Euler step with $\frac{dy}{dx} = 1x + y$, $x_0 = 0$, $y_0 = 1$, and $h = 0.5$.

Answer:

$y_1 = 1.5$.

Working:

Euler's method uses $y_{n+1} = y_n + hf(x_n, y_n)$.

Here $f(x,y) = x + y$.

$f(0, 1) = 1(0) + 1 = 1$.

$y_1 = 1 + 0.5(1) = 1.5$.

S2

Complete the Euler approximations step by step.

Use Euler's method with $\frac{dy}{dx} = 2x + y$, $y(0) = 1$, and $h = 0.2$ to estimate $y(0.6)$.

Answer:

$y(0.6) \approx 1.984$.

Working:

Euler's method uses $y_{n+1} = y_n + hf(x_n, y_n)$.

Step 1: $f(0, 1) = 2(0) + 1 = 1$.

$y_1 = 1 + 0.2(1) = 1.2$.

Step 2: $f(0.2, 1.2) = 2(0.2) + 1.2 = 1.6$.

$y_2 = 1.2 + 0.2(1.6) = 1.52$.

Step 3: $f(0.4, 1.52) = 2(0.4) + 1.52 = 2.32$.

$y_3 = 1.52 + 0.2(2.32) = 1.984$.

S3

Use the Euler update to find the missing step size.

An Euler step gives $y_1 = 1.5$ from $y_0 = 1$ when $f(x_0, y_0) = 2$. Find h.

Answer:

$h = 0.25$.

Working:

Euler's update is $y_1 = y_0 + hf(x_0, y_0)$.

$1.5 = 1 + h(2)$.

$0.5 = 2h$.

$h = \frac{0.5}{2} = 0.25$.

Teacher answers & working

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S4

Use one Euler step with $y_{n+1} = y_n + hf(x_n, y_n)$.

Use one Euler step with $\frac{dy}{dx} = 3x + y$, $x_0 = 0$, $y_0 = 2$, and $h = 0.5$.

Answer:

$y_1 = 3.$

Working:

Euler's method uses $y_{n+1} = y_n + hf(x_n, y_n)$.

Here $f(x,y) = 3x + y$.

$f(0, 2) = 3(0) + 2 = 2.$

$y_1 = 2 + 0.5(2) = 3.$

EA1

Complete the Euler approximations step by step.

Use Euler's method with $\frac{dy}{dx} = 1x + y$, $y(0) = 1$, and $h = 0.25$ to estimate $y(0.75)$.

Answer:

$y(0.75) \approx 2.15625.$

Working:

Euler's method uses $y_{n+1} = y_n + hf(x_n, y_n)$.

Step 1: $f(0, 1) = 1(0) + 1 = 1$.

$y_1 = 1 + 0.25(1) = 1.25.$

Step 2: $f(0.25, 1.25) = 1(0.25) + 1.25 = 1.5$.

$y_2 = 1.25 + 0.25(1.5) = 1.625.$

Step 3: $f(0.5, 1.625) = 1(0.5) + 1.625 = 2.125$.

$y_3 = 1.625 + 0.25(2.125) = 2.15625.$

EA2

Use the Euler update to find the missing step size.

An Euler step gives $y_1 = 4$ from $y_0 = 2$ when $f(x_0, y_0) = 8$. Find h.

Answer:

$h = 0.25.$

Working:

Euler's update is $y_1 = y_0 + hf(x_0, y_0)$.

$4 = 2 + h(8).$

$2 = 8h.$

$h = \frac{2}{8} = 0.25.$

Teacher answers & working

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EB1

Use one Euler step with $y_{n+1} = y_n + hf(x_n, y_n)$.

Use one Euler step with $\frac{dy}{dx} = 2x + y$, $x_0 = 0$, $y_0 = 1$, and $h = 0.25$.

Answer:

$y_1 = 1.25$.

Working:

Euler's method uses $y_{n+1} = y_n + hf(x_n, y_n)$.

Here $f(x,y) = 2x + y$.

$f(0, 1) = 2(0) + 1 = 1$.

$y_1 = 1 + 0.25(1) = 1.25$.